

CHANGDE, China

WMO: 576620

Lat: 29.1168N

Lon: 111.6770E

Elev: 497

StdP: 14.43

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.8	31.8	14.9	11.9	40.1	19.7	15.1	40.8	14.4	40.4	12.3	41.5	3.7	0	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	97.6	79.8	95.4	79.7	93.2	79.2	82.9	91.6	81.9	90.7	80.9	89.5	5.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.8	162.8	88.1	79.7	156.9	87.1	78.7	151.6	86.2	47.5	92.0	46.3	90.8	45.2	89.7	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.5	12.0	10.0	27.4	100.6	2.8	2.0	25.4	102.0	23.8	103.2	22.2	104.3	20.1	105.7		
(5)			25.4	84.6	2.4	1.3	23.7	85.6	22.3	86.3	21.0	87.1	19.3	88.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.4	41.7	45.9	54.5	64.7	73.0	78.9	84.8	83.6	76.0	66.3	55.9	46.1		
	DBStd	15.99	6.08	7.57	8.00	7.67	6.00	5.42	5.32	5.54	6.26	6.30	7.00	5.66		
	HDD50	636	264	155	43	2	0	0	0	0	0	0	25	147		
	HDD65	2627	721	535	338	96	7	0	0	0	3	60	280	587		
	CDD50	5888	8	40	183	444	714	866	1078	1040	781	507	203	24		
	CDD65	2408	0	1	13	88	257	416	613	575	334	102	9	0		
	CDH74	24486	0	8	82	584	1866	3946	7794	6945	2766	464	31	0		
	CDH80	10832	0	1	20	165	622	1540	4011	3375	1007	88	3	0		
(14) Wind	WSAvg	4.4	3.7	4.1	4.7	4.6	4.6	4.5	4.7	4.9	4.7	4.2	3.9	3.8		
(15) Precipitation	PrecAvg	54.2	2.3	3.0	4.1	6.5	7.3	8.1	7.7	4.5	3.0	3.3	3.1	1.4		
	PrecMax	80.7	5.4	6.8	10.3	15.8	14.3	16.1	16.7	11.5	13.5	6.5	7.2	3.5		
	PrecMin	30.6	0.7	0.3	1.6	1.9	3.5	2.6	1.3	0.6	0.3	0.6	0.2	0.0		
	PrecStd	9.7	1.2	1.8	1.9	3.0	2.6	3.5	3.7	2.5	2.6	1.8	1.8	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.3	72.2	82.3	89.4	93.0	96.4	100.1	100.4	96.6	86.8	77.9	65.9		
		MCWB	51.7	57.9	66.5	70.8	73.1	78.9	80.5	79.0	79.0	70.7	62.6	53.2		
	2%	DB	57.4	65.1	75.1	84.9	89.7	93.5	97.9	97.6	91.7	82.0	72.8	61.1		
		MCWB	48.1	54.3	61.4	69.5	73.1	78.4	80.1	79.7	77.6	68.2	60.8	50.4		
	5%	DB	53.6	60.3	70.5	80.7	86.6	90.8	96.0	95.0	88.5	78.9	69.0	57.4		
		MCWB	46.3	51.3	59.1	67.5	72.3	77.7	80.0	80.3	76.4	65.8	59.1	48.7		
	10%	DB	50.6	56.6	66.4	76.7	83.3	87.9	93.8	92.3	85.7	76.0	65.8	54.4		
		MCWB	44.6	49.4	57.6	65.8	71.0	76.8	79.8	79.9	74.6	65.2	57.7	47.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.0	60.9	69.8	75.7	78.7	82.2	84.3	84.5	81.4	74.9	66.1	55.5		
		MCDB	58.9	70.4	79.3	84.9	87.7	90.5	93.4	92.2	90.4	81.2	72.2	60.9		
	2%	WB	49.9	55.7	64.6	72.1	76.6	80.6	82.9	83.1	79.8	71.0	63.0	52.5		
		MCDB	55.6	61.5	71.1	80.6	84.2	88.5	92.1	91.3	88.2	78.0	69.5	58.6		
	5%	WB	47.5	52.7	61.3	69.5	74.8	79.4	82.0	82.0	78.4	68.9	60.9	50.2		
		MCDB	52.6	58.5	68.2	76.8	82.1	87.4	91.1	90.6	86.5	75.7	66.7	56.0		
	10%	WB	45.0	50.4	58.6	67.0	73.2	78.2	81.0	80.9	76.7	67.0	58.9	48.1		
		MCDB	49.8	56.0	64.6	74.8	80.1	85.7	89.9	89.4	84.1	73.5	64.2	53.4		
(35) Mean Daily Temperature Range	MDBR		9.2	10.3	11.7	12.6	12.6	11.4	11.8	11.9	11.9	11.5	11.0	10.3		
	5% DB	MCDBR	16.4	18.4	19.6	19.7	18.5	15.7	15.1	15.4	15.8	16.7	17.0	17.5		
		MCWBR	9.3	9.7	9.5	8.4	6.8	5.3	4.4	4.6	5.7	6.2	7.3	9.0		
	5% WB	MCDBR	14.9	15.6	17.1	17.0	14.5	13.5	13.8	13.7	14.1	13.9	13.8	15.0		
		MCWBR	9.1	9.0	9.6	9.2	7.0	5.7	5.2	5.4	6.5	6.7	6.9	8.5		
(40) Clear-Sky Solar Irradiance	taub		0.697	0.744	0.767	0.752	0.696	0.696	0.616	0.686	0.728	0.750	0.714	0.685		
	taud		1.465	1.422	1.414	1.483	1.621	1.645	1.842	1.661	1.554	1.463	1.494	1.506		
	Ebn at Noon		168	176	186	197	210	209	226	208	192	173	163	162		
	Edh at Noon		82	93	99	95	83	81	66	79	85	88	79	76		
(44) All-Sky Solar Radiation	RadAvg		552	645	919	1120	1224	1295	1582	1411	1143	878	717	584		
	RadStd		99	168	82	137	175	169	165	131	132	135	116	115		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	-3.42	N/A	-0.76	-0.99	N/A	N/A	N/A	N/A
Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page